Exercise7

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```r
library(dplyr) # A Grammar of Data Manipulation
library(ggplot2) # Create Elegant Data Visualisations Using the Grammar of Graphics
library(kableExtra) # Construct Complex Table with 'kable' and Pipe Syntax
library(mlogit) # Multinomial Logit Models
```

```r
data("Heating",
     package = "mlogit")
```

```r
H <- mlogit.data(Heating,
                 shape = "wide",
                 choice = "depvar",
                 varying = c(3:12))
```

1. Restate the blue bus-red bus situation as a nested logit model. What are the

marginal and conditional probabilities of this model?

difine the aggregates utilities of the bus alternatives: $e^{IV_{bus}}$

Then the probabilities of choosing car:

$$P_{car} = \frac{e^{V_{car}}}{e^{V_{car}} + e^{IV_{bus}}}$$

The probabilities of choosing car:

$$P_{bus} = \frac{e^{IV_{bus}}}{e^{V_{car}} + e^{IV_{bus}}}$$

Conditional Probabilities

Probability of Blue Bus in the Bus Nest:

$$P_{bb|bus} = \frac{e^{V_{bb}/\lambda}}{e^{V_{bb}/\lambda} + e^{V_{rb}/\lambda}}$$

Probability of Red Bus in the Bus Nest:

$$P_{rb|bus} = \frac{e^{V_{rb}/\lambda}}{e^{V_{bb}/\lambda} + e^{V_{rb}/\lambda}}$$

Mariginal Probabilities

For the blue bus:

$$P_{bb} = P_{bus} * P_{bb|bus}$$

For the red bus:

$$P_{rb} = P_{bus} * P_{rb|bus}$$

2. Usemodel `nl2` in this chapter and calculate the direct-point elasticity at the mean

values of the variables, for an increase in the installation costs of Gas Central systems.

```r
nl2 <- mlogit(depvar ~ ic + oc, H,
              nests = list(room = c( 'er', 'gr'), central = c('ec', 'gc', 'hp')),
              # this will set all lambdas to be the same
              un.nest.el = TRUE,
              steptol = 1e-12)
summary(nl2)
```

```
##
## Call:
## mlogit(formula = depvar ~ ic + oc, data = H, nests = list(room = c("er",
##     "gr"), central = c("ec", "gc", "hp")), un.nest.el = TRUE,
##     steptol = 1e-12)
##
## Frequencies of alternatives:choice
##       ec       er       gc       gr       hp
## 0.071111 0.093333 0.636667 0.143333 0.055556
##
## bfgs method
## 19 iterations, 0h:0m:0s
## g'(-H)^-1g = 0.0443
## successive function values within tolerance limits
##
## Coefficients :
##                   Estimate  Std. Error  z-value  Pr(>|z|)
## (Intercept):er -1.1290e+00  7.8672e-02 -14.3508 < 2.2e-16 ***
## (Intercept):gc -1.0474e-02  1.5358e-02  -0.6820 0.4952521
## (Intercept):gr -1.1817e+00  8.0048e-02 -14.7628 < 2.2e-16 ***
## (Intercept):hp -4.5459e-02  1.5516e-02  -2.9298 0.0033921 **
## ic             -8.3899e-05  2.5826e-05  -3.2486 0.0011598 **
## oc             -2.2841e-04  5.9509e-05  -3.8383 0.0001239 ***
## iv              2.7374e-02  3.2338e-03   8.4649 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-Likelihood: -1003.5
## McFadden R^2:  0.018329
## Likelihood ratio test : chisq = 37.473 (p.value = 3.6549e-08)
```

3. Use model `nl2` in this chapter and calculate the cross-point elasticity at the mean

values of the variables, for a 1% increase in the operation costs of Gas Central systems.

4. Re-estimate the nested logit model in this chapter, but change the nests to types

of energy as follows: - Gas: gas central, gas room. - Electricity: electric central, electric room, heat pump. Use a single coefficient for the inclusive variables (i.e., set un.nest.el = TRUE). Are the results reasonable? Discuss